Software Requirements Specification

Explore Nepal Travel & Tourism Website

Version 1.0

Prepared by: Explore Nepal Group

Institution:

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Document Control

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1.0	9 Oct 2026	Explore Nepal Project Team	Initial draft based on project summary and supplied template.

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1. Introduction

1.1 Purpose

This document explains the main requirements for Explore Nepal, a travel and tourism website about Nepal. It describes what visitors can do on the website, how the system should work, and the basic quality and security requirements. Project details will be finalized by the team during design and implementation.

1.2 Document Conventions

The words “shall” and “must” describe required features. “Should” describes a recommended feature, while “may” means a feature is optional. FR numbers identify functional requirements, and NFR numbers identify nonfunctional requirements. [ASSUMPTION] means something still needs confirmation, and means a decision has not been made.

1.3 Intended Audience and Reading Suggestions

This document is for the project team, professor, and anyone helping design, build, or test the website. Sections 1 and 2 introduce the project, Sections 3 and 4 describe the interface and features, and Section 5 covers quality and security. The appendices include diagrams, key terms, and decisions that still need to be made.

1.4 Product Scope

Explore Nepal is a website where people can learn about places to visit in Nepal. Visitors will be able to browse and search destinations, read destination information and suggested itineraries, and send travel questions through an inquiry form. Administrators will manage the website content and check submitted inquiries.

For the first version, the website is planned as an information site with a travel inquiry form. It will not confirm bookings or process payments. Live availability, reviews, visitor accounts, and connections to booking services will only be added if the team agrees to include them.

1.5 References

- User-supplied “Software Requirements Specification” template (IEEE-style / Karl E. Wiegers template), 8 pages.
- Project brief: “Travel and tourism website”; description “explore Nepal”; technology stack “python”.
- IEEE 29148 requirements engineering guidance may be consulted if required by the course; the exact edition and citation should be confirmed if required by the course.

No external API, design guide, or regulatory reference was specified.

2. Overall Description

2.1 Product Perspective

Explore Nepal will be developed as a web application. Visitors will use it through a web browser, while the Python backend will handle requests and work with the database. The database will store destination information and travel inquiries. Administrators must sign in before using management features. The backend will use the Django web framework. The database and hosting service will be selected by the project team.

Component	Responsibility

Visitor browser / UI	Render listings, search, destination pages, forms, and confirmations.
Python web application	Route requests, validate input, apply rules, enforce permissions, and coordinate data access.
Database	Persist destinations, itineraries, inquiries, and administrator account data.
Admin interface	Manage destination and trip content and review inquiries.
Optional email service	Send inquiry notifications/confirmations if enabled; provider will be selected if email notifications are enabled.

2.2 Product Functions

- Display a home page and curated Nepal destinations.
- Browse destinations by category, region, activity, or approved attributes.
- Search and filter destination records.
- Display destination details, highlights, practical information, and related itineraries.
- Accept and validate travel inquiries.
- Store inquiries and show confirmation; optionally notify administrators.
- Authenticate administrators and enforce permissions.
- Create, edit, publish, unpublish, and archive destination and itinerary content.
- Review inquiries and update inquiry status.
- Provide responsive layouts, accessible labels, and clear error messages.

2.3 User Classes and Characteristics

User class	Characteristics / permissions
Visitor / tourist	Primary user; usually anonymous; browses, searches, views content, submits inquiries.
Administrator / content manager	Authenticated trusted user; manages content and inquiries.
Tour operator / tourism partner	[ASSUMPTION] May receive/respond to inquiries if role is approved; permissions to be finalized by the project team.
Developer / maintainer	Maintains code, deployment, backups, and security updates.

2.4 Operating Environment

- Client: modern desktop and mobile browsers; current versions of major desktop and mobile browsers.
- Server: the server environment must support the selected Python runtime and framework.
- Backend: Python; framework will be selected by the team.
- Database: relational database recommended; engine will be selected by the team.
- Production network traffic shall use HTTPS.

2.5 Design and Implementation Constraints

- Python is the only confirmed implementation technology; framework, versions, ORM, and database will be selected by the team before implementation.
- Separate presentation, business logic, and persistence where practical.
- Production secrets must not be committed to GitHub.
- Travel content must be reviewed by the designated content owner; content should be reviewed regularly by the designated content owner.
- Applicable privacy obligations and institutional policies must be identified before production release.
- Student-project scope may limit live booking, payments, and third-party integrations.

2.6 User Documentation

- Visitor FAQ/help page covering search, destination pages, and inquiry submission.
- Administrator guide for sign-in, content publishing, and inquiry management.
- GitHub README with setup, environment variables, database initialization, tests, and deployment.
- Documentation will be written in English for version 1.0.

2.7 Assumptions and Dependencies

- [ASSUMPTION] Destination and itinerary information will be supplied or approved by the team.
- [ASSUMPTION] Core browsing works without visitor accounts.
- [ASSUMPTION] An inquiry is not a binding booking and does not guarantee availability or price.
- [ASSUMPTION] Administrators are provisioned by the team; public admin registration is not allowed.
- Framework, database, hosting, email provider, domain, analytics, and external API choices.
- Whether favorites, reviews, accounts, maps, live booking, or payments are included.
- Content languages, expected traffic, retention period, and support ownership.

3. External Interface Requirements

3.1 User Interfaces

The website should have easy-to-follow navigation, clear page titles, destination cards, search and filter options, and a footer with contact and policy links. Pages should work on both computers and mobile phones. Forms should show which fields are required and explain how to correct any errors.

- Public pages: Home, Destinations, Destination Details, Itineraries/Guides, About, Contact/Inquiry, FAQ, Privacy Notice.
- Admin pages: Dashboard, Destination Management, Itinerary Management, Inquiry List, Inquiry Details, Admin Login.
- Messages shall be actionable and shall not expose stack traces or internal system details.
- Branding, colors, wireframes, and supported languages will be finalized during design.

3.2 Hardware Interfaces

The website does not need special hardware. Users only need a device, a supported web browser, and an internet connection. Server requirements can be decided after the expected number of users is known.

3.3 Software Interfaces

Interface	Data exchanged	Purpose / status
Browser ↔ Python app	HTTP requests; HTML or JSON responses	Primary visitor/admin interaction; HTTPS in production.
Python app ↔ database	Queries/transactions for content, users, inquiries	Database engine/schema to be finalized by the project team.
App ↔ email provider	Notification message and delivery status	Optional dependency; provider will be selected if email notifications are enabled.
GitHub / CI	Source, dependency manifests, tests, docs	Repository URL and CI workflow to be finalized by the project team.

3.4 Communications Interfaces

The team can use HTTP while developing locally, but the deployed website should use HTTPS. The server must check form data before saving it. If email notifications are added, the email service must be

configured securely, and passwords or keys must not be placed in the GitHub repository. Other email settings will be decided if this feature is included.

4. System Features

4.1 Destination Browsing and Discovery

Priority: High. Visitors can open the home page or destination list and select a destination to read more. If no destinations have been added, the website should explain that the list is empty.

Flow: The visitor opens the destination page, chooses a destination, and reads its details. If no destinations are available, the website displays an explanatory message.

Functional Requirements:

- FR-01: The system shall display a home page with navigation and featured Nepal destinations.
- FR-02: The system shall show only published destination records to public visitors.
- FR-03: A destination page shall show its name, region, description, highlights, images if available, and configured practical information.
- FR-04: The system shall show a helpful empty-state message when no destinations are available.
- FR-05: Related itineraries/destinations shall be shown when relationships exist; otherwise the section may be omitted.

4.2 Search and Filtering

Priority: High. Visitors can search by keyword and use available filters to narrow the destination list.

Flow: The visitor enters a keyword or selects a filter. The website shows matching results and lets the visitor clear the search if needed.

Functional Requirements:

- FR-06: The system shall search approved destination fields such as name, region, and description.
- FR-07: The system shall provide only filters supported by stored data, such as region, activity, or destination type.
- FR-08: The system shall combine active filters consistently and show the active filter state.
- FR-09: The system shall show a no-results message and a way to modify or clear search criteria.
- FR-10: The system shall safely handle empty, long, or malformed search input without exposing internal errors.

4.3 Itineraries and Travel Guides

Priority: Medium. If travel guides are included, visitors can read the suggested route, duration, activities, and other available notes.

Flow: The visitor selects a published guide and the website displays the information that has been added.

Functional Requirements:

- FR-11: The system shall list published itineraries or guides if the module is enabled.
- FR-12: An itinerary page shall show title, duration, route/stops, description, and practical notes when supplied.
- FR-13: Suggested itineraries shall be labeled informational and shall not imply live availability or guaranteed pricing.

4.4 Travel Inquiry Submission

Priority: High. Visitors can send a travel question through a form. The website checks the information, saves valid inquiries, and displays a confirmation.

Flow: The visitor completes and submits the form. The website checks the required fields, saves the inquiry if the information is valid, and confirms that it was received.

Functional Requirements:

- FR-14: The system shall provide an inquiry form with configurable contact fields and a message field.
- FR-15: The system shall validate required fields and formats on client and server sides.
- FR-16: The system shall reject invalid submissions with field-specific messages and shall not store them as valid inquiries.
- FR-17: The system shall store accepted inquiries with a unique identifier, timestamp, and initial status.
- FR-18: After storage, the system shall confirm receipt and state that the inquiry is not a confirmed booking.
- FR-19: The system should notify an administrator of new inquiries if email notification is configured.
- FR-20: The system shall implement basic anti-spam protection appropriate to the framework and hosting.

4.5 Administrator Authentication and Authorization

Priority: High. Administrators must sign in before they can manage website content or view inquiries.

Flow: The administrator enters their login details. The website checks them and allows access only when they are valid.

Functional Requirements:

- FR-21: The system shall require authentication before allowing administrative actions.
- FR-22: Passwords shall use a modern salted hash provided by a trusted framework/library; plaintext passwords shall never be stored.
- FR-23: The system shall deny unauthorized access to protected routes and data.
- FR-24: The system shall provide logout that invalidates the active session.
- FR-25: The system shall not permit public self-registration for administrator privileges.

4.6 Content Management

Priority: High. Signed-in administrators can add or edit destination information and choose whether it is visible to visitors.

Flow: The administrator edits a destination or guide, checks the required information, and publishes or hides the content.

Functional Requirements:

- FR-26: An administrator shall create, edit, publish, unpublish, and archive destination records.
- FR-27: An administrator shall create and edit itinerary/guide records if the module is enabled.
- FR-28: The system shall validate required content fields before publishing.
- FR-29: The system should record creation/update timestamps and responsible admin identity.
- FR-30: Unpublished or archived content shall not appear in public listings or search results.

4.7 Inquiry Management

Priority: Medium. Administrators can review travel inquiries and update their status as they respond to them.

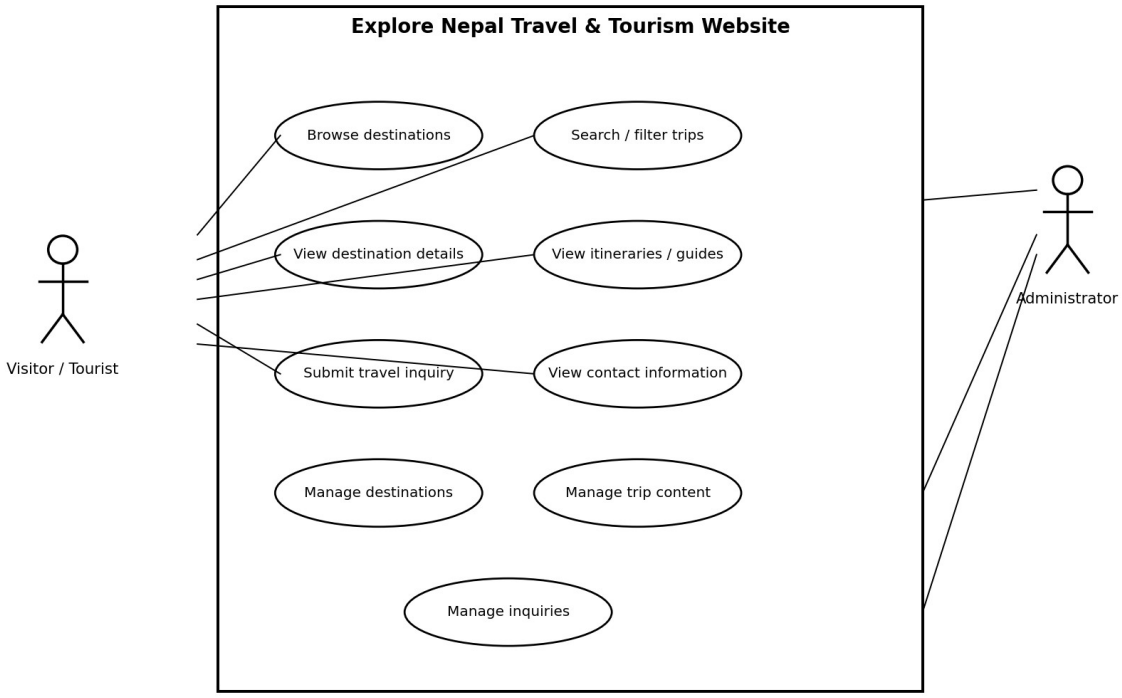
Flow: The administrator opens an inquiry, reads the message, and changes its status when needed.

Functional Requirements:

- FR-31: An administrator shall list and inspect submitted inquiries.
- FR-32: An administrator shall update inquiry status using a configured set, such as New, In Progress, Responded, or Closed.
- FR-33: Inquiry details shall be restricted to authorized users and not exposed on public pages.
- FR-34: The system should support filtering inquiries by date, status, or reference identifier.

4.8 Use Case Diagram

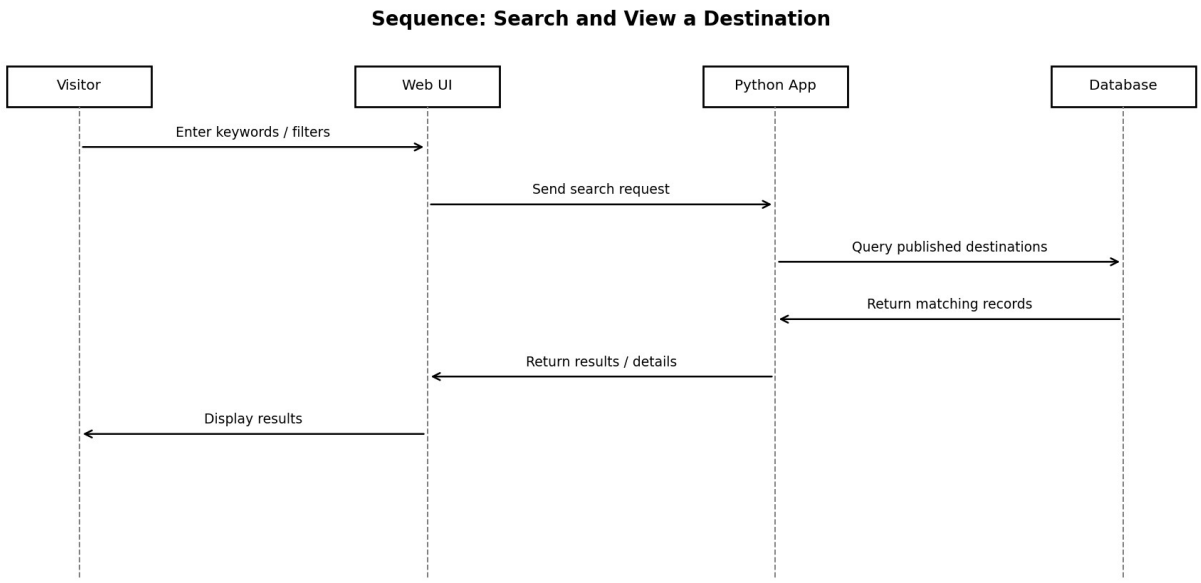
The use case diagram shows how visitors and administrators use the website. A separate tour-operator role is not confirmed and should only be included if the team decides it is needed.



Administrator workflow is assumed; confirm role and scope.

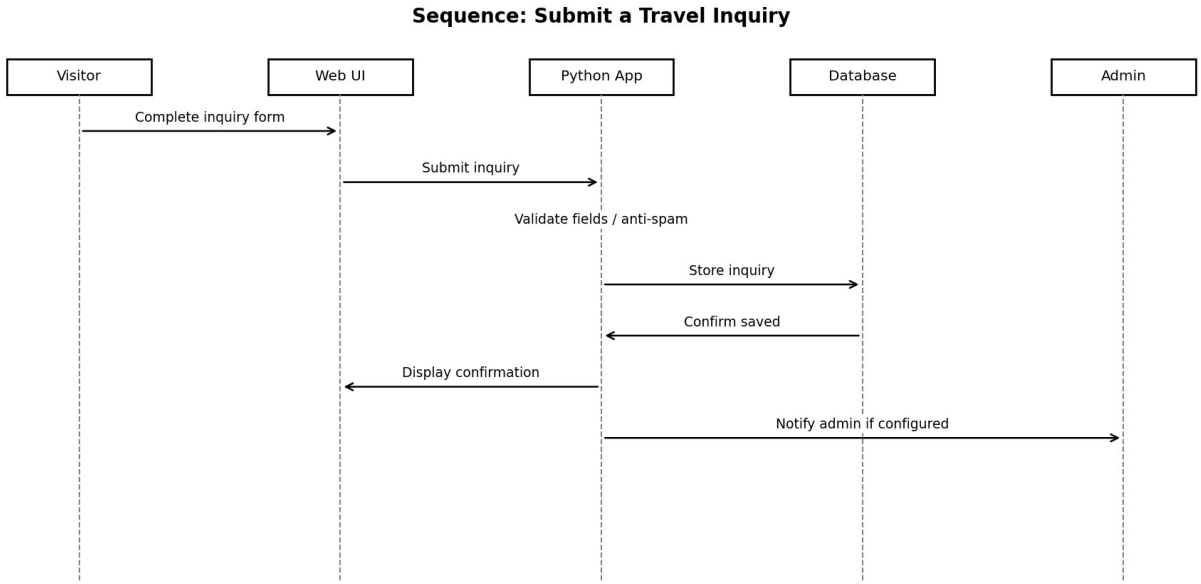
4.9 Sequence Diagram — Search and View Destination

This diagram shows how a visitor searches for a destination and opens a result. If nothing matches, the website displays a message and lets the visitor try again.



4.10 Sequence Diagram — Submit Travel Inquiry

This diagram shows the steps for submitting a travel inquiry. The server checks the form and saves valid information. Email notification will only happen if the team adds that feature.



5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-01: Under a reference load of 50 concurrent visitors, 95% of ordinary page requests should complete within 3 seconds, excluding third-party delays. [TARGET; validate in test environment].
- NFR-02: Search should return results within 2 seconds for up to 10,000 destination/itinerary records in the agreed test environment. [TARGET].
- NFR-03: Images shall be optimized; long lists should use pagination or incremental loading.

- NFR-04: Long-running form submissions should show progress/disabled-submit state and prevent duplicate submissions where feasible.

5.2 Safety Requirements

- NFR-05: Travel information shall be labeled as subject to change; visitors should verify current conditions, permits, and official guidance before travel.
- NFR-06: An inquiry shall not be presented as a confirmed booking, ticket, permit, or service guarantee.
- NFR-07: Safety guidance, if shown, shall be reviewed by the content owner and include a last-reviewed date where feasible. The content owner will review safety guidance before publication.

5.3 Security Requirements

- NFR-08: Production traffic shall use HTTPS.
- NFR-09: Server-side validation and parameterized database access or a trusted ORM shall be used.
- NFR-10: The application shall apply output escaping and protections against common vulnerabilities such as XSS and CSRF where applicable.
- NFR-11: Admin routes and inquiry records shall require authorization; visitors shall not access private inquiries.
- NFR-12: Secrets and production values shall be stored outside source code and excluded from version control.
- NFR-13: Only personal data needed to answer an inquiry shall be collected; a privacy notice shall explain purpose and contact method.
- NFR-14: Admin access should be logged without recording passwords or unnecessary sensitive form contents.
- NFR-15: Inquiry data retention and deletion rules shall be defined before production. .

5.4 Software Quality Attributes

- NFR-16 Usability: In testing, a first-time visitor should reach a destination page from the home page within three navigation actions.
- NFR-17 Accessibility: Use semantic headings, keyboard-operable controls, image alternatives, and form labels. Accessibility practices will follow semantic headings, keyboard-operable controls, image alternatives, and form labels.
- NFR-18 Compatibility: Core functions shall work in agreed current desktop/mobile browsers; the team will agree on supported current desktop and mobile browsers during testing.
- NFR-19 Maintainability: Separate routes/controllers, business logic, data access, templates/static assets, and configuration where practical.
- NFR-20 Reliability: Log server errors with timestamps and show friendly error pages instead of raw stack traces.
- NFR-21 Testability: Automated tests should cover search, validation, authorization, publication state, and inquiry storage.
- NFR-22 Portability: Environment-based configuration shall support local execution and deployment without hard-coded machine-specific paths.

5.5 Business Rules

- BR-01: Only published destinations and guides are visible to anonymous visitors.
- BR-02: Only authorized administrators may manage content or inspect inquiry details.
- BR-03: An inquiry is a request for follow-up, not a confirmed reservation or contract.
- BR-04: Required fields and accepted values shall be validated before records are saved.

- BR-05: Content accuracy and update frequency are the responsibility of the designated content owner; the designated content owner will review content regularly.
- BR-06: A tour-operator account/assignment workflow is optional and requires approval before implementation.

6. Other Requirements

6.1 Data Requirements

Expected entities include Destination, Itinerary/Guide, Inquiry, Administrator/User, and optional Media/Category. Field types, relationships, deletion rules, and migrations will be finalized during database design. Content records should have a stable identifier, title/name, publication state, and timestamps. Inquiry records should include a reference, timestamp, contact information, message, and status.

6.2 Backup and Recovery

Before production use, define backup frequency, retention, restore procedure, and responsible owner. Backup frequency, retention, restore procedure, and responsibility will be defined before production use. Uploaded images and external content files must be included in backup planning.

6.3 Localization and Content

The website interface and content for version 1.0 will be in English. Version 1.0 uses English; additional languages and localization may be considered in a future version.

6.4 Deployment and Repository

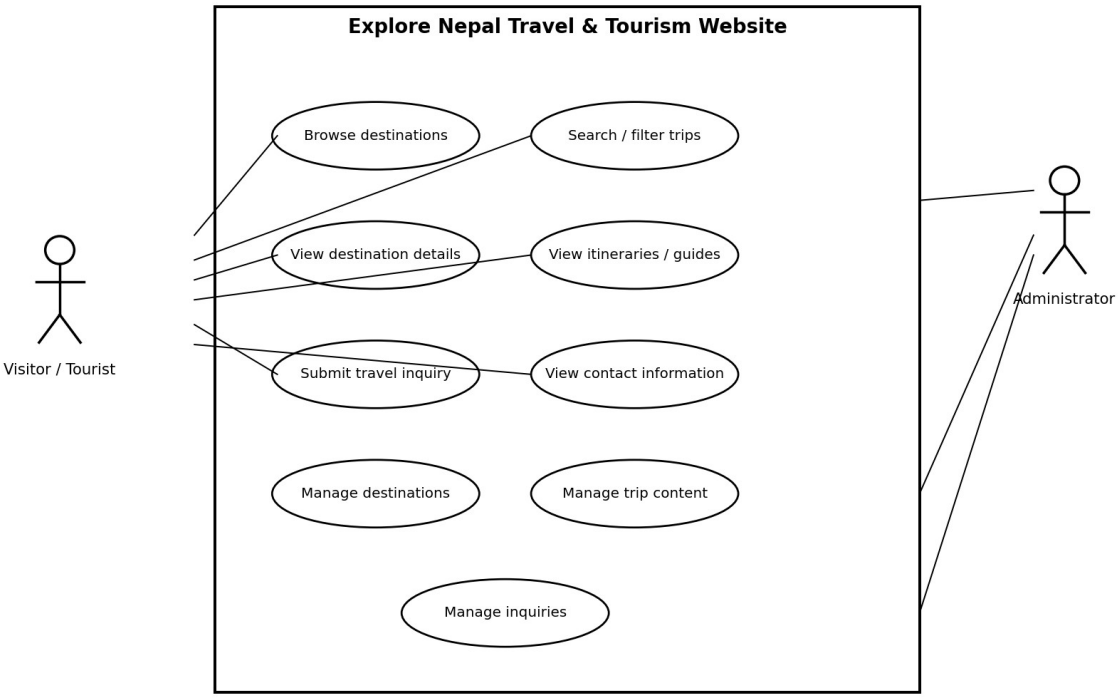
The GitHub repository should include source code, dependency manifest, schema/migrations, tests, README, example environment configuration with placeholders, and this SRS. It must not include credentials, API keys, production data, or private inquiry exports. The team will document the repository location and release process in the project README.

Appendix A: Glossary

Term	Definition
SRS	Software Requirements Specification.
Visitor	User of public site features, usually without signing in.
Administrator	Authorized user who manages content and inquiries.
Destination	A Nepal location or attraction represented by a content record.
Itinerary	Suggested sequence of destinations or activities.
Inquiry	Visitor request for information; not a confirmed booking.
Functional requirement	Capability or behavior the system must provide.
Nonfunctional requirement	Quality, constraint, or operational requirement.
to be finalized by the project team	To be determined; a decision or value not yet supplied.

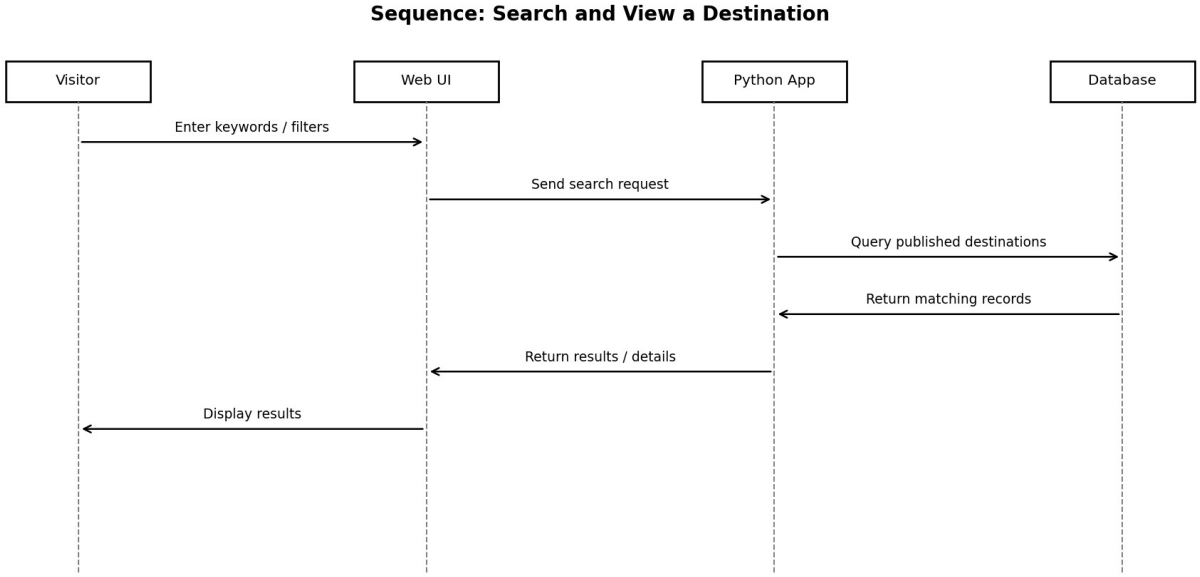
Appendix B: Analysis Models

B.1 Use case diagram

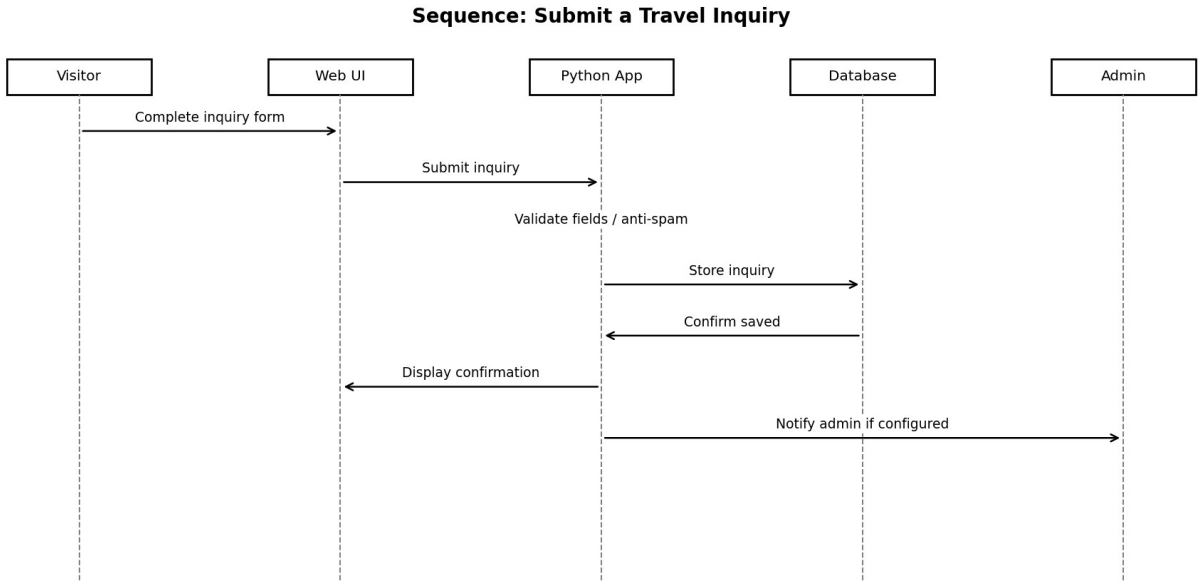


Administrator workflow is assumed; confirm role and scope.

B.2 Sequence diagram — Search and view destination



B.3 Sequence diagram — Submit travel inquiry



System Context Model

The context model shows the Explore Nepal website as the main system and identifies the people or services that interact with it. Visitors use the website to find information about Nepal and send inquiries. Administrators update destination information and manage inquiries. Map and email services are shown as optional because the team has not confirmed whether they will be used.

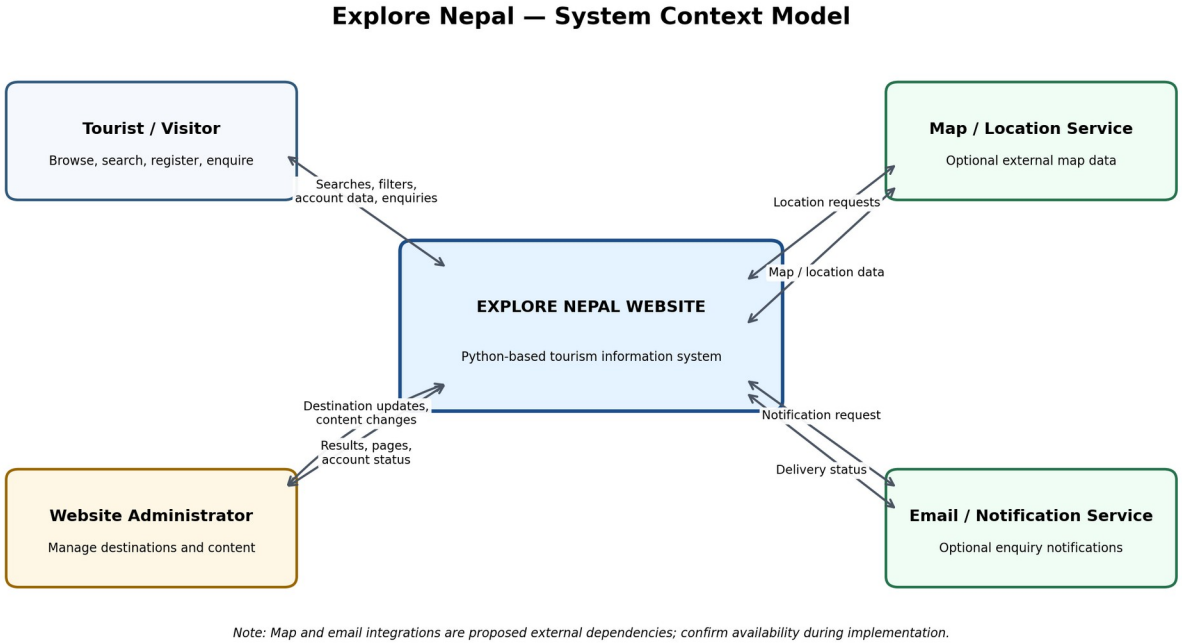


Figure: Explore Nepal System Context Diagram

External Entities and Information Flows

External entity	Information sent to the website	Information received from the

		website
Tourist / Visitor	Search terms, filters, registration/login details, enquiries	Destination details, search results, pages, account status, enquiry acknowledgement
Website Administrator	Destination records, images, article/content edits, moderation actions	Admin pages, validation messages, saved/updated status
Map / Location Service (optional)	Map tiles, coordinates, or location information	Location/map requests, if the integration is implemented
Email / Notification Service (optional)	Delivery status or error response	Notification request for an enquiry or account event, if implemented

Context Model Assumptions and Constraints

- The Explore Nepal website is the system boundary; external map and email providers are outside that boundary.
- The Python technology choice is specified by the project brief; the specific Python framework and hosting platform remain to be finalized by the project team.
- Map/location and email/notification integrations are optional proposals, not confirmed implemented features.
- Online booking, payment processing, and live availability are not included unless the team and professor explicitly approve them.

Items to Confirm (to be finalized by the project team)

- Confirm whether the website will use an external map/location API.
- Confirm whether enquiry notifications will be sent by email and select a provider if required.
- Confirm whether user registration and login are in the final project scope.